

SuperStore Sales Performance Analysis

Analytical report on sales, profit, and discount performance from 2011 to 2014

\$12.6M

Sales

\$1.47M

Profit

11.6%

Profit margin

24.5%

Loss-making rows

51,290

Rows

Executive Summary

Global Superstore sales grew steadily from \$2.26M in 2011 to \$4.30M in 2014. Profit also increased, but overall profitability remains moderate at an 11.6% profit margin. The main business risk is not demand volume; it is discount discipline. Nearly one quarter of transaction rows are loss-making, and discounts above 20% systematically push orders into negative margin territory.

Business question	Which markets, categories, and discount policies generate profit, and where does the company lose money?
Analysis period	2011–2014
Data scope	25,035 orders, 795 customers, 7 markets
Main risk	High discounts and concentrated losses in specific countries and sub-categories
Key recommendation	Cap discounts above 20% and review the Tables sub-category strategy

1. Sales and Profit Trend

Sales increased by 90.3% between 2011 and 2014, from \$2.26M to \$4.30M. Profit grew from \$249K to \$504K over the same period. However, profit did not scale as efficiently as revenue, which suggests that discounting and loss-making categories limit operating leverage.

Sales and Profit Trend, 2011-2014

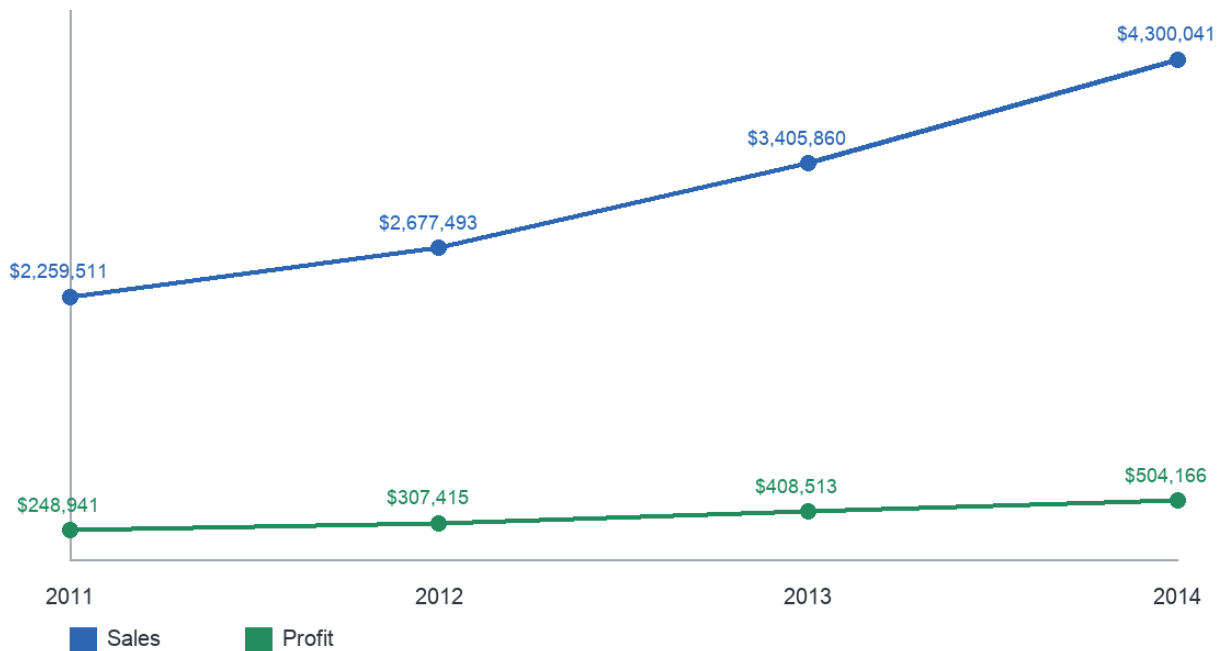


Figure 1: Sales and profit by year

Year	Sales	Profit	Orders	Profit margin
2011	\$2,259,511	\$248,941	4,440	11.0%
2012	\$2,677,493	\$307,415	5,343	11.5%
2013	\$3,405,860	\$408,513	6,721	12.0%
2014	\$4,300,041	\$504,166	8,531	11.7%

2. Market Performance

Profit is concentrated in APAC, EU, and the US. These markets provide the largest absolute profit contribution. Canada has the strongest margin at 26.6%, but its sales base is too small to act as the primary growth driver. EMEA is the weakest market by profitability, generating only \$44K in profit at a 5.4% margin.

Market	Sales	Profit	Avg. discount	Profit margin
APAC	\$3,585,833	\$437,578	14.9%	12.2%
EU	\$2,938,139	\$372,830	10.3%	12.7%
US	\$2,297,354	\$286,397	15.6%	12.5%
LATAM	\$2,164,687	\$221,643	13.6%	10.2%
Africa	\$783,776	\$88,872	15.7%	11.3%
EMEA	\$806,184	\$43,898	19.6%	5.4%
Canada	\$66,932	\$17,817	0.0%	26.6%

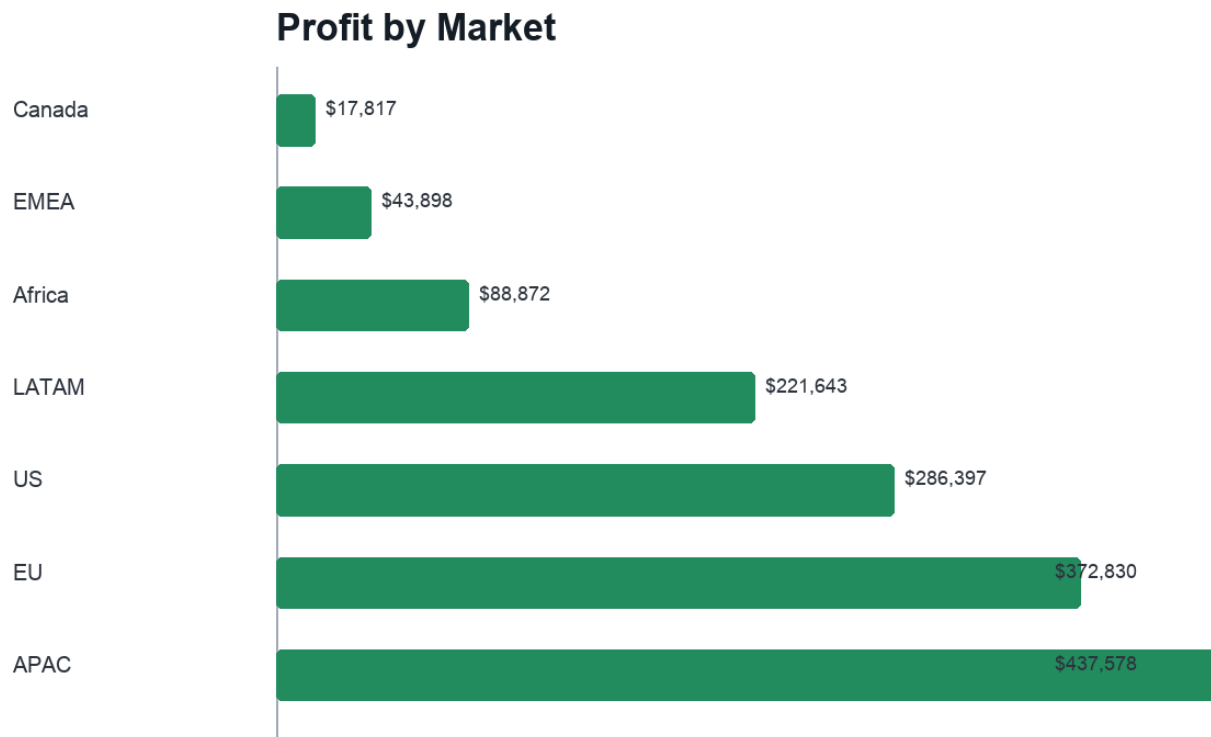


Figure 2: Profit by market

3. Category Performance

Technology and Office Supplies perform similarly, with profit margins of 14.0% and 13.7%. Furniture is materially weaker at a 7.0% margin. The main issue inside Furniture is the Tables sub-category, which generated a \$64K loss.

Category	Sales	Profit	Avg. discount	Profit margin
Technology	\$4,744,691	\$663,779	13.5%	14.0%
Office Supplies	\$3,787,330	\$518,474	13.7%	13.7%
Furniture	\$4,110,884	\$286,782	16.8%	7.0%

4. Discount Impact

The strongest finding is the relationship between discounts and profitability. Orders with no discount have a 25.3% profit margin. At 20–30% discounts, margin already turns negative at -5.5%. At discounts above 50%, losses become severe, with a -111.0% profit margin.

Discount bucket	Sales	Profit	Orders	Profit margin
0%	\$6,992,734	\$1,770,695	15,211	25.3%
0–10%	\$1,962,633	\$339,767	3,028	17.3%
10–20%	\$1,757,296	\$173,255	4,363	9.9%
20–30%	\$382,540	-\$21,156	843	-5.5%
30–50%	\$1,176,078	-\$380,945	3,674	-32.4%
50%+	\$371,624	-\$412,582	2,369	-111.0%

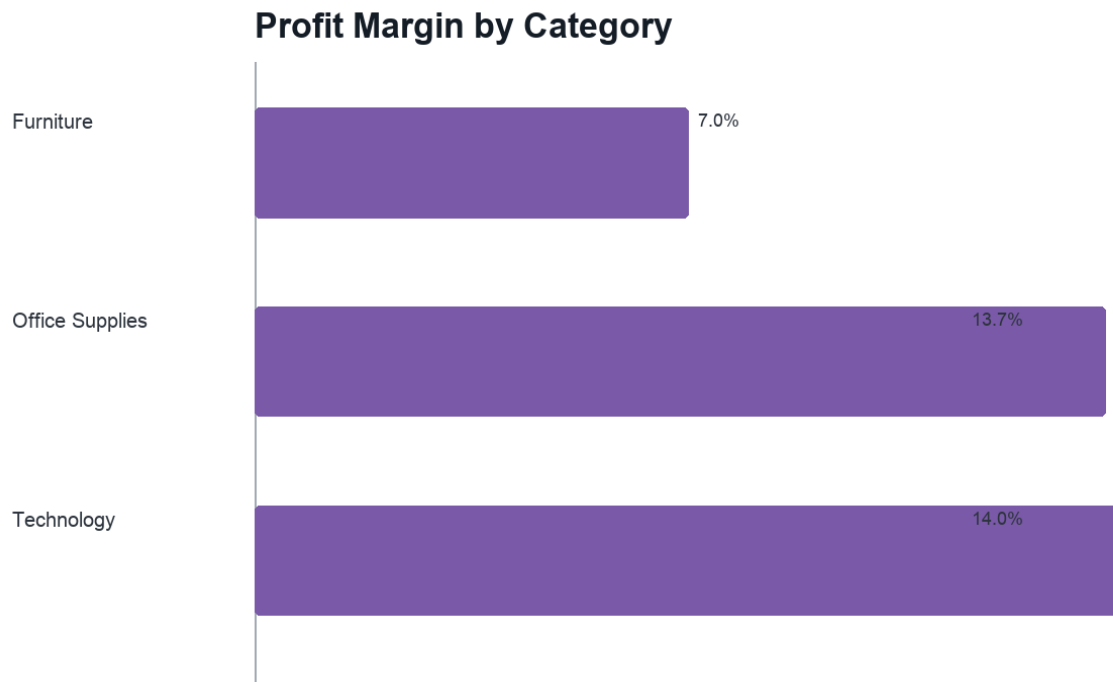


Figure 3: Profit margin by category

Discount impact by category

The overall discount chart includes all markets, countries, categories, segments, and years. The category-level view confirms that the discount pattern is broad-based: every major category becomes loss-making once discounts reach the 20–30% range. Furniture is the weakest category before heavy discounts as well, with only a 4.6% margin at 10–20% discounts.

Discount impact by market

The market-level view shows that the same discount threshold is visible across most major markets. APAC, EU, US, and LATAM are profitable up to the 10–20% bucket, but each turns negative at 20–30%. Africa and EMEA have fewer intermediate discount buckets in the dataset, but their 50%+ discount buckets show severe losses. Canada has no discounted sales in this dataset, so only the 0% bucket is available.

5. Loss Concentration by Country

Losses are concentrated in countries with high average discounts. Turkey, Nigeria, and the Netherlands are among the largest negative-profit countries. This points to a commercial policy issue, not just a local demand or operations issue.

Country	Sales	Profit	Avg. discount	Profit margin
Turkey	\$108,524	-\$98,447	60.0%	-90.7%
Nigeria	\$54,347	-\$80,751	70.0%	-148.6%
Netherlands	\$77,520	-\$41,070	48.2%	-53.0%
Honduras	\$90,137	-\$29,482	40.7%	-32.7%
Pakistan	\$58,873	-\$22,447	44.5%	-38.1%

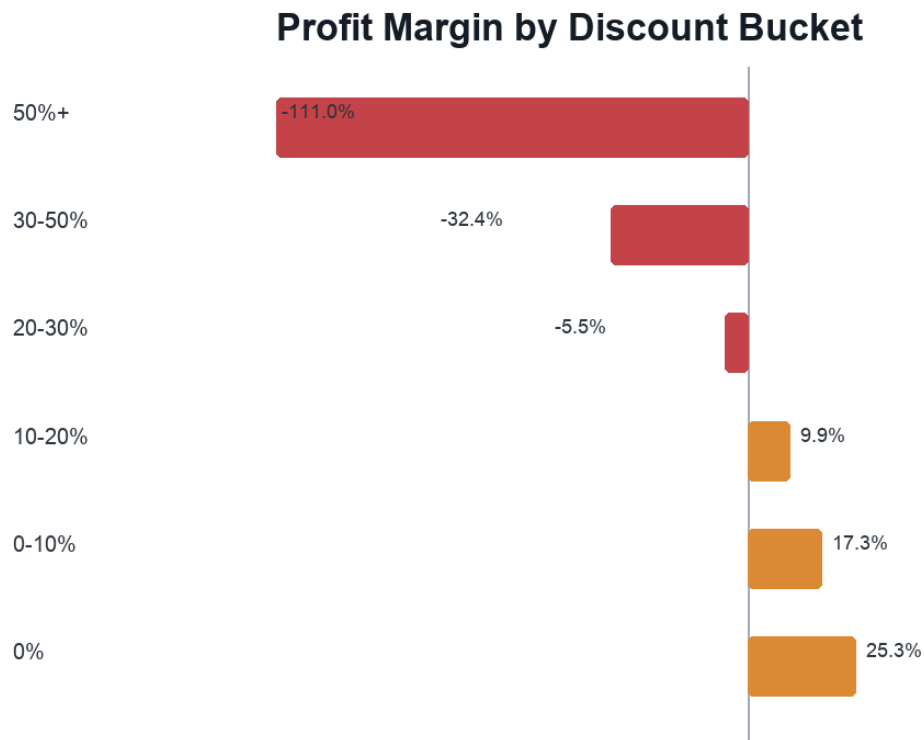


Figure 4: Profit margin by discount bucket

Recommendations

- Cap discounts above 20%, especially in countries and categories with negative margins.
- Review the Tables strategy, including pricing, discount rules, logistics, and product mix.
- Scale proven practices from APAC, EU, and the US because these markets generate the largest absolute profit.
- Audit EMEA, Turkey, and Nigeria at the pricing and promotional-policy level.
- Use Canada as a profitability benchmark, but not as the main growth driver because of its small sales base.

Appendix: Reproducibility

The analysis can be reproduced with:

```
python scripts/analyze_superstore.py
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Discount Impact by Category

Cell values show profit margin. Green is profitable; red is loss-making.

	0%	0-10%	10-20%	20-30%	30-50%	50%+
Furniture	24.2%	16.7%	4.6%	-6.2%	-32.0%	-106.3%
Office Supplies	25.6%	15.9%	13.6%	-6.8%	-36.1%	-113.2%
Technology	25.9%	18.9%	12.0%	-1.9%	-30.3%	-115.3%

Figure 5: Profit margin by discount bucket and category

Discount Impact by Market

Cell values show profit margin. Green is profitable; red is loss-making.

	0%	0-10%	10-20%	20-30%	30-50%	50%+
APAC	24.3%	16.5%	9.8%	-3.7%	-32.3%	-92.3%
EU	25.2%	16.6%	9.0%	-12.4%	-41.3%	-95.1%
US	29.5%	16.6%	11.6%	-10.0%	-24.8%	-119.2%
LATAM	23.9%	22.2%	6.8%	-3.7%	-28.0%	-109.1%
Africa	24.7%	14.0%	n/a	n/a	n/a	-148.3%
EMEA	24.6%	n/a	n/a	n/a	-4.8%	-98.0%
Canada	26.6%	n/a	n/a	n/a	n/a	n/a

Figure 6: Profit margin by discount bucket and market